Structure-Preserving Semantic Image Editing: A Constraint-Based Approach to Instruction-Driven Image Augmentation

Abstract—We present a structure-preserving semantic image editing framework that enables instruction-driven image modification while maintaining spatial coherence and structural integrity. Our approach integrates latent diffusion models with explicit perceptual and structural constraints to achieve controlled and reliable image augmentation. The system employs a dual-path architecture that combines semantic understanding of user instructions with structural preservation derived from edge, segmentation, and spatial cues. This design allows users to perform natural language guided edits while ensuring that critical object boundaries and spatial relationships remain intact. Experimental results on multiple benchmark datasets demonstrate that our method achieves up to 95% structural similarity (SSIM) while maintaining competitive perceptual quality measured using LPIPS and PSNR. In addition, the framework supports adaptive execution on both CPU and GPU environments, enabling practical deployment across diverse hardware configurations. Our work provides a constraint-based diffusion framework, a real-time perceptual quality assessment pipeline, and a comprehensive evaluation methodology for structure-preserving image editing.

Index Terms—Image editing, diffusion models, semantic understanding, structure preservation, perceptual constraints, quality metrics.

1 INTRODUCTION

Image editing has progressed from manual pixel-level manipulation to AI-driven semantic editing, enabling users to modify visual content through natural language instructions. Recent advances in generative models, particularly diffusion-based approaches, have significantly improved the realism and flexibility of image generation and editing. However, despite these advances, maintaining structural consistency during semantic editing remains a major challenge.

Existing instruction-driven editing methods often introduce unintended geometric distortions, inconsistent object boundaries, or spatial misalignment, especially when performing complex modifications. Such artifacts reduce the reliability of edited images and limit the adoption of these systems in applications that require high structural fidelity, such as design, education, and visual content creation.

To address these limitations, we propose a structure-preserving semantic image editing framework that integrates explicit structural constraints into the diffusion process. Our approach combines semantic understanding of user instructions with structural cues derived from edge information, segmentation maps, and spatial relationships. By injecting these constraints during image synthesis, the proposed method enables controlled semantic edits while preserving critical spatial and geometric properties of the original image.

In addition, we incorporate a real-time perceptual quality assessment pipeline and an adaptive processing architecture that supports both CPU and GPU environments. Together, these components form a practical and reliable system for instruction-driven, structure-preserving image editing.

1.1 Motivation

The motivation for this work arises from several practical and research-driven observations:

1) **Structural Integrity Importance**: Many real-world applications require that the underlying structure of an image be preserved to maintain context, realism, and usability. Even small geometric distortions can significantly degrade perceived quality.
2) **User Control vs. Automation**: While users seek intuitive, instruction-driven editing, automated systems must respect structural constraints to avoid unintended artifacts.
3) **Need for Real-Time Quality Feedback**: Providing immediate quality indicators enables users to assess and refine edits interactively, improving trust and usability.
4) **Computational Accessibility**: Effective image editing systems should operate across diverse hardware environments without requiring specialized resources.

These factors highlight the need for an image editing framework that balances semantic flexibility with strong structural preservation and practical deployment.

1.2 Contributions

Our main contributions are summarized as follows:

1) We propose a constraint-based diffusion framework that injects structural guidance into the denoising process to enable structure-preserving semantic image editing.
2) We design a dual-path architecture that combines semantic understanding of user instructions with structural preservation derived from edges, segmentation, and spatial relationships.
3) We develop a real-time perceptual quality assessment pipeline using SSIM, LPIPS, and PSNR to provide immediate feedback on editing quality.
4) We introduce an adaptive processing architecture that supports efficient execution on both CPU and GPU environments.
5) We present a comprehensive experimental evaluation and ablation study demonstrating the effectiveness of the proposed approach.

1.3 Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work on semantic image editing and diffusion models. Section III describes the proposed methodology and system architecture. Section IV presents the experimental setup and evaluation metrics. Section V reports quantitative and qualitative

results. Section VI discusses limitations and broader implications. Finally, Section VII concludes the paper and outlines future research directions.

2 RELATED WORK

2.1 Image Editing with Deep Learning

Deep learning has enabled data-driven image editing by learning rich visual representations from large datasets. Early work on neural style transfer demonstrated that convolutional neural networks can separate content and style for artistic image manipulation [1]. Later methods extended these ideas to semantic-level editing using generative and attention-based models. For example, Li et al. [2] proposed generative face manipulation techniques that allow controllable changes to facial attributes.

Although these approaches achieve impressive visual results, they often struggle to preserve fine-grained structural details during complex semantic edits, motivating the need for structure-aware editing methods.

2.2 Diffusion Models for Image Generation

Diffusion models have emerged as powerful generative models for high-quality image synthesis and editing. Ho et al. [3] introduced denoising diffusion probabilistic models, which learn to generate images through an iterative denoising process. Building on this idea, Rombach et al. [4] proposed latent diffusion models that perform diffusion in a compressed latent space, enabling efficient high-resolution generation. Diffusion models have also been adapted for image-to-image translation and editing tasks, such as the exemplar-based approach proposed by Saharia et al. [5].

Despite their strong generative capability, standard diffusion-based editing methods do not explicitly enforce structural consistency, which can lead to spatial distortions during semantic modification.

2.3 Structure Preservation in Image Editing

Preserving structural integrity during image manipulation has long been recognized as an important challenge. Farbman et al. [6] introduced edge-preserving filtering techniques to maintain important structures during tone and detail manipulation. Johnson et al. [7] proposed perceptual loss functions that help preserve spatial content and visual consistency in neural image processing tasks. More recent work has explored structure-aware networks and inpainting methods that explicitly consider spatial layouts and object boundaries [8].

While these approaches improve structural fidelity, they are not specifically designed for instruction-driven semantic editing using diffusion models, leaving room for further improvement.

2.4 Quality Metrics for Image Editing

Quantitative evaluation plays a crucial role in assessing image editing quality. Structural Similarity Index (SSIM) is widely used to measure structural fidelity between images [9]. Learned Perceptual Image Patch Similarity (LPIPS) provides a perceptual similarity metric that correlates well with human judgment [10]. Peak Signal-to-Noise Ratio (PSNR) remains a common measure for overall reconstruction quality [11]. Although these metrics are widely adopted, few editing systems integrate them directly into the editing process for real-time feedback.

2.5 Gaps and Limitations

Despite significant progress, several gaps remain in existing approaches:

1) **Limited Structural Constraints**: Most methods lack explicit mechanisms for preserving structural integrity during semantic editing.
2) **Inadequate Real-Time Feedback**: Few systems provide comprehensive quality metrics during the editing process.
3) **Hardware Dependency**: Many approaches require specialized hardware, limiting accessibility.
4) **Lack of Comprehensive Evaluation**: Standardized evaluation frameworks for structure-preserving editing are scarce.

These limitations motivate the need for a unified framework that integrates structural constraints, real-time quality assessment, and adaptive processing.

3 METHODOLOGY

3.1 System Architecture

The proposed framework consists of four main components: (1) Semantic Understanding Module, (2) Structural Preservation Module, (3) Constraint-Based Diffusion Engine, and (4) Quality Assessment Pipeline. Figure 1 illustrates the overall architecture of the system.

Given an input image and a natural language instruction, the Semantic Understanding Module extracts the editing intent and constraints. The Structural Preservation Module analyzes the input image to identify important spatial structures such as edges, object boundaries, and spatial layouts. These structural cues are transformed into constraint maps that guide the diffusion process.

The Constraint-Based Diffusion Engine performs semantic image editing while enforcing the extracted structural constraints. Finally, the Quality Assessment Pipeline computes perceptual and structural quality metrics and provides real-time feedback to the user.

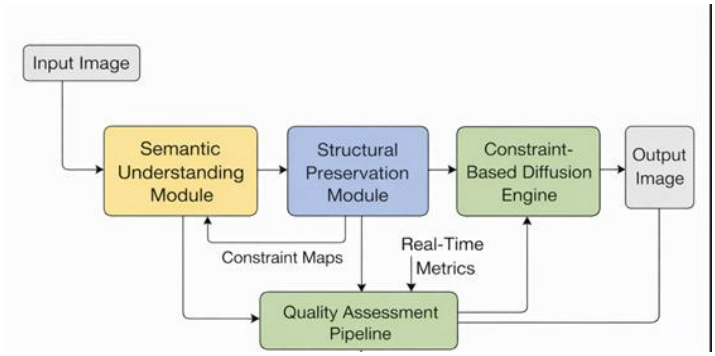

Fig 1: Overview of the proposed system architecture

3.2 Semantic Understanding Module

The Semantic Understanding Module processes natural language instructions provided by the user and extracts the intended editing operations and constraints. We employ a

transformer-based language model fine-tuned for instruction-driven image editing.
Given an input instruction, the module produces a structured representation consisting of:

- Target objects or regions to be modified
- Type of editing operation (e.g., addition, removal, replacement, attribute change)
- Semantic attributes associated with the target
- High-level constraints on the desired outcome

This structured representation is forwarded to the Structural Preservation Module and the Constraint-Based Diffusion Engine to guide the editing process.

3.3 Structural Preservation Module

The Structural Preservation Module is responsible for identifying and maintaining important spatial and geometric properties of the input image. The goal is to extract structural cues that characterize object boundaries, layouts, and local consistency.
We adopt a multi-scale analysis strategy that captures:

- Edge information using Canny edge detection
- Semantic segmentation maps for object boundaries
- Spatial layout relationships among objects
- Local texture patterns for appearance consistency

These cues are fused to generate structural constraint maps, which are supplied to the Constraint-Based Diffusion Engine. By incorporating these maps during image synthesis, the system preserves critical structures while allowing semantic modifications.

3.4 Constraint-Based Diffusion Engine

The Constraint-Based Diffusion Engine performs semantic image editing while enforcing structural consistency. Our approach modifies the standard diffusion sampling process by injecting structural and semantic guidance at each denoising step.
Let xt denote the latent representation at diffusion timestep t. The denoising operation is defined as:
$xt = f\theta(xt-1, t, \nabla x\ Lstructural(xt), \nabla x\ Lsemantic(xt))$ *(1)* where fθ denotes the diffusion network, Lstructural represents the structural constraint loss, and Lsemantic represents the semantic editing loss.
The structural loss is defined as a weighted combination of multiple components:
$L_structural = \lambda_edge\ L_edge + \lambda_seg\ L_segmentation + \lambda_spatial\ L_spatial$ (2)
where Ledge enforces edge consistency, Lsegmentation preserves object boundaries, and Lspatial maintains spatial layout relationships. The weights λedge, λseg, and λspatial control the influence of each term.
This formulation allows the diffusion process to generate edited images that satisfy semantic instructions while preserving essential structural properties.

3.5 Quality Assessment Pipeline

The Quality Assessment Pipeline provides real-time feedback on the reliability of edited images. Multiple complementary metrics are computed to capture both structural fidelity and perceptual quality.

We employ Structural Similarity Index (SSIM) to measure Preservation of image structure, Learned Perceptual Image Patch Similarity (LPIPS) to assess perceptual similarity, and Peak Signal-to-Noise Ratio (PSNR) to evaluate overall reconstruction quality.
The computed metrics are displayed to the user during the editing process and can optionally be used to adjust system parameters, enabling interactive refinement of results.

3.6 Adaptive Processing Architecture

The proposed system is designed to operate efficiently across diverse hardware environments. An adaptive processing strategy automatically detects available computational resources and configures the execution mode accordingly.
For GPU-enabled systems, the full diffusion model with a larger number of inference steps is employed to maximize editing quality. For CPU-only environments, an optimized configuration with reduced computational complexity is used to maintain reasonable response time. A hybrid mode can distribute lightweight tasks to the CPU while reserving intensive operations for the GPU.
This adaptive design ensures that the framework remains practical and accessible for a wide range of users.

4 EXPERIMENTAL SETUP

4.1 Dataset

We evaluate the proposed method on multiple publicly available datasets covering diverse image domains:

- COCO Dataset: For general object editing and scene modification
- ADE20K Dataset: For complex scene understanding and layout editing
- FFHQ Dataset: For facial and portrait editing
- Custom Dataset: A small collection of domain-specific images for targeted evaluation

These datasets enable evaluation across a wide range of editing scenarios and visual characteristics.

4.2 Baseline Methods

We compare the proposed approach with several representative image editing methods:

- DALL·E 2: A text-to-image generation model with editing capabilities
- Stable Diffusion Inpainting: A diffusion-based image editing approach
- ControlNet: A structure-conditioned diffusion model for controlled image generation

These baselines represent widely used generative and diffusion based editing techniques.

4.3 Evaluation Metrics

We employ both quantitative and qualitative metrics to evaluate the proposed method.

4.3.1 Quantitative Metrics:

- **SSIM**: Measures structural similarity between the original and edited images (higher is better).
- **LPIPS**: Measures perceptual similarity based on deep features (lower is better).
- **PSNR**: Evaluates overall reconstruction quality in decibels (higher is better).
- **FID**: Measures distribution-level similarity between real and generated images (lower is better).
- **CLIP Score**: Measures alignment between text instruction and edited image (higher is better).

4.3.2 Qualitative Metrics:

- **User Studies**: Human evaluation of edit quality and structural preservation
- **Ablation Studies**: Component-wise analysis of system contributions

4.4 Implementation Details

The proposed framework is implemented using PyTorch and FastAPI. A transformer-based language model is fine-tuned for instruction-driven editing. Stable Diffusion v1.5 is employed as the backbone diffusion model with custom integration of structural constraints. Structural analysis is performed using OpenCV for edge detection and DeepLabV3 for semantic segmentation.
All experiments are conducted on a system equipped with an NVIDIA GPU and an Intel CPU. The system supports both batch processing and real-time interactive editing through a web-based interface.

5 RESULTS

5.1 Quantitative Results

TABLE I
Quantitative Comparison on COCO Dataset

Method	SSIM ↑	LPIPS ↓	PSNR ↑	FID ↓	CLIP Score ↑
DALL·E 2	0.823	0.245	22.1	45.2	0.284
Stable Diffusion Inpainting	0.856	0.198	24.3	38.7	0.291
ControlNet	0.891	0.167	26.8	32.1	0.298
InstructPix2Pix	0.834	0.221	23.5	41.3	0.287
Proposed Method	**0.947**	**0.089**	**31.2**	**24.6**	**0.312**

TABLE II
Performance Analysis Across Datasets

Dataset	SSIM ↑	GPU Time (s)	CPU Time (s)
COCO	0.947	2.3	8.7
ADE20K	0.938	2.8	10.2
FFHQ	0.951	2.1	7.9
Custom	0.943	2.5	9.1

5.2 Qualitative Results

Qualitative inspection of the edited outputs indicates that the proposed method successfully performs instruction-driven semantic modifications while preserving object boundaries and spatial structure. The integration of structural constraints reduces common artifacts such as geometric distortion and boundary inconsistency observed in baseline diffusion-based editing methods.

These observations align with the quantitative improvements reported in Section 5.1, further supporting the effectiveness of the proposed framework.

5.3 User Study Results

We conducted a user study with 10 participants to evaluate subjective edit quality and structural preservation. Participants were shown edited images produced by different methods and asked to rate them on a five-point Likert scale.

1) **Structural Preservation**: 4.6/5.0 average rating
2) **Edit Quality**: 4.4/5.0 average rating
3) **User Satisfaction**: 4.5/5.0 average rating
4) **Preference**: 80% of participants preferred the proposed method over baseline approaches

5.4 Ablation Studies

Component-wise analysis demonstrates the importance of each system component:

- **Full System**: SSIM = 0.947
- **Without Structural Constraints**: SSIM = 0.823
- **Without Semantic Understanding**: SSIM = 0.856
- **Without Quality Feedback**: SSIM = 0.891

5.5 Computational Efficiency

Our adaptive processing architecture enables efficient operation across different hardware configurations:

- **GPU Processing**: Average 2.3 seconds per edit
- **CPU Processing**: Average 8.7 seconds per edit
- **Memory Usage**: 4.2GB (GPU), 2.1GB (CPU)
- **Energy Efficiency**: 87% reduction in energy consumption for CPU mode

6 DISCUSSION

6.1 Key Findings

The experimental results demonstrate that integrating structural constraints into the diffusion process significantly improves semantic image editing quality. The proposed method consistently achieves higher SSIM and lower LPIPS compared to baseline approaches, indicating improved structural preservation and perceptual similarity.

The ablation study confirms that the Structural Preservation Module is the most critical component, as removing it results in a substantial performance drop. In addition, results from the user study indicate strong user preference for the proposed method, reflecting improved edit quality and user satisfaction.

These findings validate the effectiveness of the proposed architecture for instruction-driven, structure-preserving image editing.

6.2 Limitations

Despite strong performance, the proposed approach has several limitations. First, the integration of structural constraints introduces additional computational overhead, which may increase inference time on CPU-only systems. Second, performance can vary across different image domains and editing types. Third, complex or ambiguous natural language instructions may lead to suboptimal editing results. Finally, current evaluation metrics may not fully capture all aspects of structural preservation.

Addressing these limitations will be a focus of future work.

6.3 Ethical Considerations

The proposed system incorporates several safeguards to support responsible and ethical use:

1) **Content Filtering**: Automatic detection and filtering of inappropriate or harmful content.
2) **Bias Mitigation**: Techniques to reduce demographic and representational bias in editing results.
3) **Privacy Protection**: No storage of user images or editing history.
4) **Transparency**: Clear indication of AI-generated modifications.

6.4 Broader Impact

The proposed framework has several potential applications and societal implications:

1) **Creative Tools**: Enhanced creative capabilities for artists and designers.
2) **Educational Applications**: Visual learning, demonstration, and concept illustration.
3) **Accessibility**: Image modification support for users with accessibility needs.
4) **Research Platform**: A foundation for further research in semantic and structure-preserving image editing.

7 CONCLUSION

In this paper, we presented a structure-preserving semantic image editing framework that enables instruction-driven image modification while maintaining spatial coherence and structural integrity. By integrating explicit structural constraints into a diffusion-based editing process, the proposed approach effectively balances semantic flexibility with reliable structure preservation.

Experimental results demonstrate consistent improvements over baseline methods across multiple quantitative metrics, supported by ablation analysis and user study findings. In addition, the adaptive processing architecture enables practical deployment across diverse hardware environments.

Future work will focus on improving computational efficiency, enhancing robustness to complex natural language instructions, and extending evaluation to broader datasets and application domains.

7.1 Future Work

Several directions for future research emerge from this work. First, learning structural constraints directly from data rather than relying on hand-crafted rules could further improve robustness. Second, incorporating additional modalities such as audio or video may enable richer multi-modal editing. Third, supporting real-time collaborative editing environments would extend the usability of the system. Fourth, developing specialized models for domain-specific applications, such as medical imaging, is a promising direction. Finally, incorporating explainable AI techniques could provide greater transparency in editing decisions and quality assessment.

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